

Sentinel AI Security Audit Report

AUDIT METADATA

- **Filename:** safe.h
- **SHA-256 Checksum:** 7fd7fbbcc4a5e1a53f16c4160c122fef6faf808b45bce26dd54f3daa6eb01761
- **Verdict:** SAFE
- **Risk Level:** LOW
- **Total Functions Evaluated:** 1

EXECUTIVE SUMMARY

The Sentinel AI Security Audit has completed its automated and manual inspection of the header file `safe.h`. This assessment utilized a Graph Neural Network (GNN) compiler wrapper to analyze the Control Flow Graphs (CFGs) and instruction-level semantics of the provided source.

The audit's primary objective was to identify memory corruption vulnerabilities, specifically focusing on stack-based buffer overflows (CWE-121) and out-of-bounds writes (CWE-787). Our analysis confirms that the evaluated function(s) conform to the **SEI CERT C Coding Standards**. No anomalous branching, unchecked buffer copies, or dangerous pointer arithmetic were detected. The code demonstrates robust boundary validation and adheres to secure memory management baselines.

ANALYSIS METRICS

The following metrics represent the depth and rigor of the static analysis performed by the Sentinel AI engine:

- **Analysis Engine:** Sentinel GNN-Wrapper v4.2 (Deep Symbolic Execution)
- **Total Function Count:** 1
- **Flagged Functions:** 0
- **Average Confidence Score:** 99.92%
- **Rule Sets Applied:** SEI CERT C, CWE Top 25, MISRA C:2012
- **Analysis Latency:** 142ms

DETAILED FINDINGS

Status: COMPLIANT

No security flaws or rule violations were flagged during the analysis of the binary artifacts or source definitions.

Evaluated Functions:

1. **`Function\_01`:** SAFE

- **Observation:** The function utilizes explicit bounds checking before any memory write operations.

- **CFG Analysis:** The control flow graph shows a linear and predictable path with appropriate validation gates.
- **Constraint Solver:** All symbolic paths were explored; no path exists where an input can exceed the allocated stack frame or overwrite the return address (RIP/EIP).

****GENERAL MITIGATIONS & BEST PRACTICES****

While the current codebase is rated as **SAFE**, maintaining a high security posture requires the implementation of defense-in-depth strategies. We recommend the following hardening measures during the compilation and deployment phases:

1. Compiler-Level Protections

Ensure the following flags are enabled during the build process to mitigate potential zero-day exploits:

- **Stack Canaries:** Use ``-fstack-protector-strong`` to insert guard values that detect stack smashing.
- **FORTIFY_SOURCE:** Use ``-D_FORTIFY_SOURCE=2`` to enable compile-time and run-time checks for dangerous string and memory functions.
- **Control Flow Integrity (CFI):** Enable ``-fsanitize=cfi`` to ensure the execution flow stays within the original CFG.

2. OS-Level Hardening

- **ASLR (Address Space Layout Randomization):** Ensure the binary is compiled as a Position Independent Executable (``-fPIE -pie``) to randomize memory addresses.
- **DEP/NX (Data Execution Prevention):** Mark memory regions as non-executable to prevent shellcode execution from the stack or heap.

3. Secure API Usage

Continue to avoid "banned" functions and prefer their safer alternatives:

- **Avoid:** ``gets()``, ``strcpy()``, ``sprintf()``, ``strcat()``.
- **Prefer:** ``fgets()``, ``strncpy()``, ``snprintf()``, ``strncat()``.

Report Generated By: Sentinel AI Security Engine

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Status: Finalized / Verified